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# INT396

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# UNSUPERVISED LEARNING

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## Lecture #0

# Course Introduction

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- **Course Code :** INT396
- **Course Title :** Unsupervised Learning
- **L-T-P Structure :** 2 0 2
- **Credits:** 03
- **Nature of Course :** Core

# Vision

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To be a globally recognized school through excellence in teaching, learning, and research, creating Computer Science professionals, leaders, and entrepreneurs of the future who contribute to society and industry for sustainable growth.

# Mission

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- M1: To provide a learning-based environment on technical concepts applied to real-life situations with measurable outcomes.
- M2: To establish connections with the industry for curriculum design, and creating internship cum career opportunities.
- M3: To address societal issues related to regional, national and global challenges through meaningful research.
- M4: To inspire graduates for pursuing lifelong learning in professional careers.
- M5: To develop leadership potential in ethically competent entrepreneurs.

# Course Outcomes (COs)

**CO1:** Explain fundamental concepts of unsupervised learning and interpret real-world problem formulations

**CO2:** Differentiate between various clustering paradigms (hard vs. soft, partition-based vs. density-based) and compare distance/similarity metrics and their impact on clustering quality. Apply supervised learning algorithms for classification and regression.

**CO3:** Apply standard unsupervised learning algorithms such as k-Means, Hierarchical Clustering, DBSCAN, GMM, PCA, t-SNE, UMAP and association rule mining techniques on real-world datasets using Python frameworks.

**CO4:** Interpret and evaluate the quality of clustering and dimensionality reduction results using validation metrics. Implement unsupervised learning methods for clustering and dimensionality reduction.

**CO5:** Integrate multiple unsupervised techniques and construct visual dashboards or project solutions. Use Python libraries (scikit-learn, TensorFlow) to perform ML experiments.

**CO6:** Develop a substantial unsupervised learning project that involves problem identification, data preprocessing, algorithm selection, model validation, and insights presentation. Analyze and visualize results for decision support and pattern discovery.

# Program Outcomes

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**PO1:** Apply knowledge of computing, mathematics, artificial intelligence, data science, and engineering fundamentals to solve complex computing problems.

**PO2:** Identify, formulate, analyze, and solve complex computing and AI-related problems using appropriate analytical techniques.

**PO3:** Design innovative software systems, AI-enabled applications, databases, and computing solutions that meet specified requirements.

**PO4:** Conduct experiments, analyze data, evaluate results, and derive meaningful conclusions for computing and AI applications.

**PO5:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

**PO6:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

## Program Outcomes (Cont.)

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**PO7:** Apply ethical principles and professional responsibilities in software engineering, cybersecurity, data handling, and AI systems.

**PO8:** Function effectively as an individual and as a member or leader in multidisciplinary and diverse teams.

**PO9:** Communicate effectively with technical and non-technical stakeholders through reports, presentations, documentation, and discussions.

**PO10:** Apply project management principles, business practices, resource planning, and financial considerations in technology-driven projects.

**PO11:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

# Program Specific Outcomes

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**PSO1:** Design, develop and deploy intelligent software systems by applying computer science fundamentals and artificial intelligence techniques including machine learning, deep learning, computer vision, natural language processing, foundation models and reinforcement learning.

**PSO2:** Analyze complex computational problems and develop scalable AI solutions using data analytics, optimization techniques, modern AI frameworks, responsible AI practices and industry-oriented engineering tools.



# Programme Education Objectives

- **PEO1:** Graduates will establish themselves as competent computing professionals by applying principles of computer science, artificial intelligence, machine learning, and emerging technologies to develop effective solutions for real-world problems.
- **PEO2:** Graduates will demonstrate innovation and entrepreneurial mindset by designing intelligent software solutions, AI-enabled applications, and data-driven systems addressing industrial and societal challenges.
- **PEO3:** Graduates will engage in lifelong learning, professional development, and upskilling to adapt to advancements in artificial intelligence, computing technologies, and evolving industry requirements.
- **PEO4:** Graduates will demonstrate professional ethics, teamwork, leadership, and responsibility while developing sustainable and trustworthy technology solutions considering societal and environmental impact.

# Course Overview & Importance

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**The AI Discovery Paradigm:** This course introduces the core paradigm of learning from unlabeled data, enabling machines to discover hidden structures, identify patterns, and generate meaningful insights without predefined target labels.

**Core Machine Learning Skills:** Master the complete unsupervised learning pipeline including similarity measures, clustering algorithms, and visualization using industry-standard Python libraries such as Scikit-learn.

**Industry Importance:** Unsupervised Learning powers today's AI applications including customer segmentation, recommendation systems, fraud and anomaly detection, pattern discovery, and business intelligence.

**Placement Readiness:** Algorithms such as K-Means, Hierarchical Clustering, DBSCAN, Local Outlier Factor (LOF), Apriori, and FP-Growth are increasingly tested in Machine Learning Engineer, Data Scientist, AI Engineer, and Analytics interviews.

# Course Content

| Unit   | Broad Topic                          | Sub-Topics & Key Concepts                                                                                                                                                                                                   | Practical Applications                                                                                                                                                                       |
|--------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit I | Foundations of Unsupervised Learning | Learning paradigms (Supervised vs. Unsupervised vs. Semi-supervised), problem formulation, exploratory data analysis, pattern discovery, similarity & distance metrics (Euclidean, Manhattan, Cosine), real-world use cases | Customer segmentation, anomaly & fraud detection, recommendation systems, social network analysis, bioinformatics, document clustering, exploratory data analysis for business intelligence. |

# Course Content

| Unit           | Broad Topic                | Sub-Topics & Key Concepts                                                                                                                                                                                        | Practical Applications                                                                                                                           |
|----------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit II</b> | Partition-Based Clustering | Clustering fundamentals, hard vs. soft clustering, K-Means, K-Means++, K-Medoids (PAM), MiniBatch K-Means, data standardization, cluster validation (WCSS, Silhouette Score, Davies-Bouldin Index, Elbow Method) | Customer segmentation, retail analytics, image segmentation, telecom user profiling, hospital patient grouping, marketing campaign optimization. |

# Course Content

| Unit     | Broad Topic                             | Sub-Topics & Key Concepts                                                                                                                                                      | Practical Applications                                                                                                                                         |
|----------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit III | Hierarchical & Density-Based Clustering | Agglomerative & Divisive Clustering, linkage methods, dendrogram interpretation, DBSCAN, $\epsilon$ -neighborhood, MinPts, noise handling, comparison of clustering algorithms | Geographic hotspot detection, crime mapping, earthquake analysis, biological taxonomy, network intrusion detection, spatial data mining, geospatial analytics. |

# Course Content

| Unit           | Broad Topic                                        | Sub-Topics & Key Concepts                                                                                                                  | Practical Applications                                                                                                                                                |
|----------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit IV</b> | Dimensionality Reduction & Representation Learning | Curse of dimensionality, Principal Component Analysis (PCA), manifold learning, t-SNE, UMAP, autoencoder intuition, feature representation | Data visualization, face recognition, handwritten digit analysis, gene expression analysis, feature extraction, image compression, high-dimensional data exploration. |

# Course Content

| Unit   | Broad Topic                                 | Sub-Topics & Key Concepts                                                                                                                          | Practical Applications                                                                                                                                        |
|--------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit V | Association Rule Mining & Anomaly Detection | Support, Confidence, Lift, Conviction, Apriori, FP-Growth, Market Basket Analysis, Isolation Forest, Local Outlier Factor (LOF), novelty detection | Product recommendation, market basket analysis, fraud detection, cybersecurity, credit card anomaly detection, predictive maintenance, healthcare monitoring. |

# Course Content

| Unit           | Broad Topic                                        | Sub-Topics & Key Concepts                                                                                                                                                | Practical Applications                                                                                                                                                                        |
|----------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit VI</b> | Evaluation & Applications of Unsupervised Learning | Internal & external clustering validation metrics, Silhouette Score, stability analysis, interpretability, ethical AI, visualization dashboards, real-world case studies | End-to-end customer analytics dashboard, AI-based business intelligence, healthcare analytics, educational data mining, smart city applications, explainable unsupervised learning solutions. |



# List of Practicals

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1. Analyze customer behavior patterns by performing exploratory data analysis and evaluating Euclidean, Manhattan, and Cosine similarity measures on an unlabeled customer dataset.
2. Develop a customer segmentation solution by implementing and comparing k-Means and k-Medoids clustering techniques on the Mall Customers dataset.
3. Design a scalable clustering model by implementing k-Means from scratch and validating its performance using the Scikit-learn framework.
4. Investigate the influence of data preprocessing by examining the effect of feature scaling and standardization on clustering quality and model convergence.

# List of Practicals

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5. Determine the optimal clustering configuration by applying Elbow Method, Silhouette Score, and Davies–Bouldin Index on a real-world marketing dataset.
6. Analyze hierarchical relationships in unlabelled data by constructing dendrograms and comparing multiple linkage strategies using Hierarchical Clustering.
7. Develop a density-based clustering solution by applying DBSCAN and evaluating the impact of  $\epsilon$  (epsilon) and MinPts parameters on cluster formation and noise detection.
8. Evaluate clustering techniques for diverse data distributions by benchmarking k-Means, Hierarchical Clustering, and DBSCAN using clustering validation metrics.

# List of Practicals

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9. Develop an efficient feature representation model by applying Principal Component Analysis (PCA) to reduce dimensionality while preserving maximum variance.
10. Compare dimensionality reduction techniques by visualizing high-dimensional datasets using PCA, t-SNE, and UMAP for exploratory data analysis.
11. Extract actionable business insights by implementing Apriori and FP-Growth algorithms for Market Basket Analysis and interpreting association rules.
12. Develop an anomaly detection framework by comparing Isolation Forest and Local Outlier Factor (LOF) for identifying unusual observations in cybersecurity or financial datasets.

# List of Practicals

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13. Assess the quality of unsupervised learning models using clustering validation metrics, visualization techniques, and stability analysis to support model selection.
14. Develop an end-to-end unsupervised learning solution by integrating data preprocessing, clustering, dimensionality reduction, anomaly detection, visualization, evaluation, and presentation of insights for a real-world application in healthcare, finance, education, agriculture, cybersecurity, or retail.

# Innovative Pedagogies

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- **Dataset Exploration:** Analyzing unlabeled datasets to identify patterns and formulate clustering strategies.
- **Case-Based Learning (CBL) + AI Playground:** Analyzing industrial applications in healthcare, finance, retail, cybersecurity, and smart cities.
- **Visual Learning using Interactive Tools:** Using Yellowbrick, Scikit-learn visualizers, TensorFlow Embedding Projector, and Plotly to understand clustering and dimensionality reduction visually.
- **Explainable AI Demonstrations:** Discussion on interpretability of clusters and dimensionality reduction results through visualization and feature analysis.

# Assessment & Evaluation Framework

**Total Course Marks: 100**

- **Attendance: 5 Marks**
- **Continuous Assessment – CA: 45 Marks**

| CA NO. | CA TYPE            | MARKS WEIGHTAGE | COMPULSORY                                  |
|--------|--------------------|-----------------|---------------------------------------------|
| CA1    | Project            | 35              | YES                                         |
| CA2    | Test               | 10              | Best one out of 2<br>shall be<br>considered |
| CA3    | Subjective<br>Test | 10              |                                             |

- **End-Term Examination - ETE (50 Marks): MCQ + Subjective Questions.**

# Assessment & Evaluation Framework

| CA No. | Type               | Course Coverage | Assessment Description                                                                                                                                                                                                               | Compulsory                            |
|--------|--------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| CA 1   | Project Assessment | Unit I -VI      | Implementation- 10 Marks<br>Report Writing - 10 Marks<br>Viva & Presentation- 10 Marks                                                                                                                                               | Yes                                   |
| CA 2   | Test Assessment-1  | Unit I- II      | A balanced mix of Multiple Choice Questions (MCQs) and subjective questions will be used to evaluate students' conceptual clarity, analytical thinking, problem-solving skills related to optimization and evolutionary computation. | Best one out of 2 shall be considered |
| CA 3   | Test Assessment-2  | Unit III- IV    | Subjective questions will be used to evaluate students' conceptual clarity, analytical thinking, problem-solving skills related to optimization and evolutionary computation.                                                        |                                       |

# Learning Resources

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**Primary Textbook:** *HANDS-ON UNSUPERVISED LEARNING USING PYTHON: HOW TO BUILD APPLIED MACHINE LEARNING SOLUTIONS FROM UNLABELED DATA* by ANKUR A. PATEL, O'REILLY

**Reference Book:** 1. “Models and Algorithms for Unsupervised Learning” — Vaibhav Verdhan (packt / O'Reilly)  
2. *Kickstart Unsupervised Machine Learning: Master Unsupervised Machine Learning Through Pattern Discovery, Clustering, and Dimensionality Reduction to Build Intelligent, Real-World Applications-* Dr. Nimrita koul, AVA Publication

**Demo Kits & Other Readings:** *scikit-learn installation docs* , *scikit-learn Preprocessing guide*

**Virtual Lab :** *Google Colab*



# Placement & Career Relevance

## Industry Ready Skills

- Clustering & Customer Segmentation
- Similarity Measures & Distance Metrics
- Dimensionality Reduction (PCA, t-SNE, UMAP)
- Association Rule Mining (Apriori & FP-Growth)
- Anomaly Detection (Isolation Forest & LOF)
- Python, Scikit-learn & Data Visualization
- Business Intelligence & Pattern Discovery

## Industry Applications

- Customer Segmentation
- Fraud & Intrusion Detection
- Recommendation Systems
- Market Basket Analysis
- Healthcare Analytics
- Image Segmentation
- Predictive Maintenance
- Smart Cities & IoT Analytics

## Career Opportunities

Machine Learning Engineer | Data Scientist | AI Engineer | Business Intelligence Analyst | Data Analyst | AI Research Associate | Recommendation System Engineer | Analytics Consultant

## Higher Education

NPTEL, Google ML Engineer, Azure AI, AWS ML, M.Tech/MS AI & Data Science

# Revenue Generation & Entrepreneurship Opportunities

## AI Products

Customer Segmentation Platform | Fraud Detection Engine | Recommendation System |  
Retail Analytics Dashboard | Predictive Maintenance Solution

## Freelancing

Business Intelligence Dashboard Development | Customer Analytics |  
Data Mining & Clustering Solutions | Anomaly Detection Projects | AI Consulting

## Startups

Retail Intelligence | Healthcare Analytics | Cybersecurity AI | FinTech Fraud Analytics |  
Recommendation Engine as a Service | EdTech Learning Analytics

## Industry Collaboration

Capstone Projects | Industry Dataset Challenges | Sponsored AI Projects |  
Consultancy in Data Analytics | Technology Transfer & Innovation

# Innovation, Research & Project Opportunities

## Research Areas

- Explainable AI
- Responsible AI
- AutoML
- Time Series Analytics
- Healthcare AI

## Publications

- IEEE
- Springer
- Scopus Journals
- Book Chapters

## Innovation

- AI Decision Support Systems
- Explainable AI Dashboards
- Business Intelligence

## Competitions

- Smart India Hackathon
- Kaggle
- Google Solution Challenge
- Patent Filing

# Industry Certifications & Skill Development

| Recommended Certifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Professional Skill Development                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>NPTEL:</b> Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati) <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a></p> <p><b>Google Cloud:</b> Data Analytics Learning Path</p> <p><b>Microsoft Learn:</b> Azure AI Fundamentals (AI-900)</p> <p><b>Oracle University:</b> Oracle AI Foundations</p> <p><b>IBM SkillsBuild:</b> Data Science &amp; AI Fundamentals</p> <p><b>AWS Skill Builder:</b> Machine Learning Foundations</p> <p><b>Simplilearn:</b> Data Science with Python</p> <p><b>Coursera:</b> Unsupervised Learning, Recommenders &amp; Reinforcement Learning (DeepLearning.AI)</p> | <p>Python Programming for Machine Learning</p> <p>Data Preprocessing &amp; Feature Engineering</p> <p>Scikit-learn Framework</p> <p>Model Evaluation &amp; Hyperparameter Tuning</p> <p>Git &amp; GitHub for ML Projects</p> <p>Streamlit Model Deployment</p> <p>Data Storytelling &amp; Technical Presentation</p> |

# Skill Attainment

## Technical Skills

Python Programming for Machine Learning  
Data Cleaning, Preprocessing & Feature Engineering  
Classification & Regression Modeling  
Ensemble Learning Techniques  
Model Validation & Performance Evaluation  
Hyperparameter Optimization  
Scikit-learn Library  
Time Series Forecasting  
Fundamentals  
Machine Learning Pipeline Development  
Basic Model Deployment using Streamlit

## Professional Skills

Analytical & Critical Thinking  
Data-Driven Decision Making  
Problem Solving  
Team Collaboration  
Technical Documentation  
Presentation & Communication Skills  
Project Management  
Ethical AI Practices

## Industry-Aligned Competencies

- Develop complete supervised machine learning solutions using industry-standard tools.
- Select and compare appropriate ML algorithms for real-world datasets.
- Optimize and evaluate predictive models using modern validation techniques.
- Build portfolio-ready AI projects suitable for internships and placements.
- Apply supervised learning techniques to solve problems in healthcare, finance, retail, cybersecurity, and manufacturing.

# Student Expectations & Academic Conduct

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## **Attendance & Participation**

- Maintain a minimum attendance as per university regulations.
- Actively participate in lectures, coding sessions, quizzes, and classroom discussions.
- Complete pre-class reading and engage in flipped learning activities where applicable.

## **Assignments & Practical Work**

- Submit assignments, lab exercises, and projects within the prescribed deadlines.
- Write clean, well-documented, and reproducible Python code.
- Demonstrate your solution during practical evaluations whenever required.

## **Academic Integrity**

- Follow ethical coding and AI practices.
- Avoid plagiarism, code copying, and unauthorized use of AI-generated submissions.
- Properly acknowledge datasets, libraries, and research references used in projects.

## **Professional Conduct**

- Maintain discipline and respect during classroom and laboratory sessions.
- Collaborate effectively during team-based activities while ensuring individual contributions.
- Develop a growth mindset by embracing feedback and continuous improvement.

# Student Guidance & Support

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## Learning Resources

Utilize the recommended **NPTEL course (Introduction to Machine Learning – IIT Madras)** and other Open Educational Resources (OERs).

Refer to prescribed textbooks, research papers, Scikit-learn documentation, and Python tutorials for deeper understanding. Practice using Google Colab, Jupyter Notebook, Kaggle, and GitHub for hands-on learning.

## Industry Certifications

Pursue certifications from **NPTEL, Google Cloud, Microsoft Learn, AWS Skill Builder, Oracle University, IBM SkillsBuild, and Coursera** to strengthen technical expertise.

## Innovation & Research

Participate in Smart India Hackathon (SIH), Kaggle competitions, research projects, innovation challenges, and patent-oriented initiatives.

Explore Explainable AI (XAI), MLOps, and Generative AI as extensions of supervised learning.

## Faculty Support & Feedback

Seek academic guidance during faculty consultation hours.

Clarify doubts through classroom interactions, LMS discussion forums, or scheduled mentoring sessions.

Regular feedback will be provided on assignments, coding practices, and project progress to support continuous improvement.

# OER Mapping (Unit I)

| Unit          | Broad Topic                            | OER Type              | Title of OER                                                                                                                                                                      | % Map ped | Source URL                                                                                |
|---------------|----------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| <b>Unit 1</b> | Foundati ons of Unsuperv ised Learning | Video Lectures + PDFs | <b>Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati)</b> – Introduction to ML, Pattern Recognition Fundamentals, Non-parametric Learning Concepts | 90%       | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a> |



# OER Mapping (Unit II)

| Unit          | Broad Topic                 | OER Type              | Title of OER                                                                                                                                                  | % Map ped | Source URL                                                                                |
|---------------|-----------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| <b>Unit 2</b> | Partition-Based Clusterin g | Video Lectures + PDFs | <b>Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati) –</b><br>Week 8: K-Means Clustering;<br>Week 9: Fuzzy K-Means Clustering | 90%       | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a> |

# OER Mapping (Unit III)

| Unit          | Broad Topic                             | OER Type              | Title of OER                                                                                                                                                                        | % Map ped | Source URL                                                                                |
|---------------|-----------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| <b>Unit 3</b> | Hierarchical & Density-Based Clustering | Video Lectures + PDFs | Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati) – Week 9: Hierarchical Agglomerative Clustering; supplemented with instructor sessions for DBSCAN | 85%       | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a> |

## OER Mapping (Unit IV)

| Unit          | Broad Topic                                          | OER Type              | Title of OER                                                                                                                                                                                                                                      | % Map ped | Source URL                                                                                |
|---------------|------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| <b>Unit 4</b> | Dimensionality Reduction and Representation Learning | Video Lectures + PDFs | <b>Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati) – Week 7: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA);</b> supplemented with instructor sessions for t-SNE, UMAP and Autoencoders | 90%       | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a> |

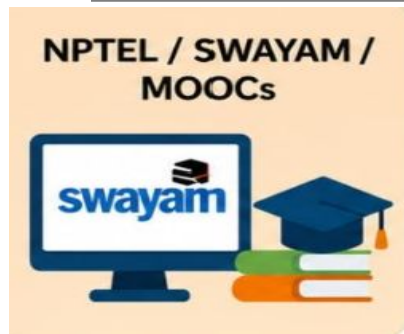
# OER Mapping (Unit V)

| Unit          | Broad Topic                                 | OER Type              | Title of OER                                                                                                                                                                                                            | % Map ped | Source URL                                                                                |
|---------------|---------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| <b>Unit 5</b> | Association Rule Mining & Anomaly Detection | Video Lectures + PDFs | <b>Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati)</b> – Pattern Discovery Concepts; supplemented with instructor-developed material for Association Rule Mining and Isolation Forest | 75%       | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a> |

# OER Mapping (Unit VI)

| Unit          | Broad Topic                                          | OER Type              | Title of OER                                                                                                                                                                                                                                     | % Map ped | Source URL                                                                                |
|---------------|------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| <b>Unit 6</b> | Evaluation and Applications of Unsupervised Learning | Video Lectures + PDFs | <b>Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati)</b> – Model Evaluation, Visualization Concepts and Real-world Machine Learning Applications; supplemented with instructor-led case studies and project work | 90%       | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a> |

# MOOC Mapped with INT396



| MOOC Details                        | Description                                                                                                                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MOOC Platform</b>                | NPTEL (SWAYAM)                                                                                                                                                                                 |
| <b>Course Name</b>                  | Machine Learning and Deep Learning – Fundamentals and Applications (IIT Guwahati)                                                                                                              |
| <b>Course Duration</b>              | 12 Weeks                                                                                                                                                                                       |
| <b>Course URL</b>                   | <a href="https://nptel.ac.in/courses/108103192">https://nptel.ac.in/courses/108103192</a>                                                                                                      |
| <b>Mapping with INT527 Syllabus</b> | 86% (Approx.)                                                                                                                                                                                  |
| <b>Eligibility for Benefits</b>     | Students must successfully complete the NPTEL course and obtain Completion Certificate.                                                                                                        |
| <b>Academic Benefits</b>            | Complete waiver of all Continuous Assessments (CA-1, CA-2, CA-3, etc.) for students who earn the Completion Certificate.<br>*Students will be awarded marks as per the university MOOC policy. |

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# Education Revolution

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# Why EDU-Revolution!!!

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- Career Readiness
- Enhance Real World Skills
- Promote Innovation and Creativity
- Foster Industry Relations
- Develop Entrepreneurial Mindset
- Increase Student Engagement and Motivation
- Promote Lifelong Learning



# Academic initiatives under EDU Revolution

**Revenue Generation**



**Projects & Hackathons**



**Internship Opportunities**  
(National/International)



**NPTEL / SWAYAM / MOOCs**



**Grade-Upgradation**



**10 % Attendance Benefits**



**Duty Leaves**



**Social Media Presence**



**Recognition of Prior Learning (RPL)**



**Semester Abroad**



**Community Service**



**Extra / Co-curricular Activities**



# EDU Revolution Benefit Mapped with INT396



## RPL PROCESS:

1. **Application:** Through UMS Portal
2. **Submission of Evidence:** Upload relevant documents such as: Coursera, NPTEL, edX, or other recognized certifications/ Project reports/ GitHub repositories/ Internship or industry experience certificates/ Workshop.
1. **Document Verification:** By Competent Authority
2. **Competency Assessment/Examination:** Upon passing in RPL Examination, Student shall be eligible for the RPL Benefit.

# EDU Revolution Benefit Mapped with INT396



## GRADE UPGRADATION PROCESS:

1. **Application:** Through UMS Portal
2. **Submission of Evidence:** Upload relevant documents such as:
  - Patent(Filed/Published)- Should only be mapped with AI /ML.
  - 1st/2nd/3rd Position in Technical Competition/Hackathons conducted by IITs/NITs/Central/State Universities.
  - Winner/Joint Winner of Smart India Hackathon 2026.
1. **Document Verification:** By Competent Authority
2. **\*Academic Benefit:** Upon successful Document Verification, Student shall be eligible for the Grade Upgradation Benefits as per the University Policy.

# EDU Revolution Benefit Mapped with INT396

## Projects & Hackathons



### PROCESS:

1. **Application:** Through UMS Portal
2. **Problem Statement:**
  - ISRO- BAH 2026- Fill in the Frames Seamlessly - Enhancing Temporal Resolution of Satellite Imagery using the techniques of AI/ML based on Optical Flow <https://hack2skill.com/event/bah2026/#challenges>
1. **Submission of Evidence:** Certificate of Completion through ISRO-BAH 2026
2. **Document Verification:** By Competent Authority
3. **\*Academic Benefit:** Upon successful Document Verification, Student shall be eligible for the Benefits as per the University Policy.

# Interfaces and Contact Information

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UMS link to apply for EDU Revolution initiatives:

***UMS Navigation --- LMS □ EDU Revolution: BE THE CHANGE***

In case of any query in this regard the student can approach for resolution through RMS at the following path:

***UMS Navigation--- Relationship management system □ EDU Revolution: BE THE CHANGE***

OR get your queries resolved through the **Helpline Number: 01824-520150.**

# Feedback

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**Feedback Mechanism:** It is mandatory for all students to participate in academic improvement processes and fill out all types of feedback regularly throughout the semester.